

Experiment 1

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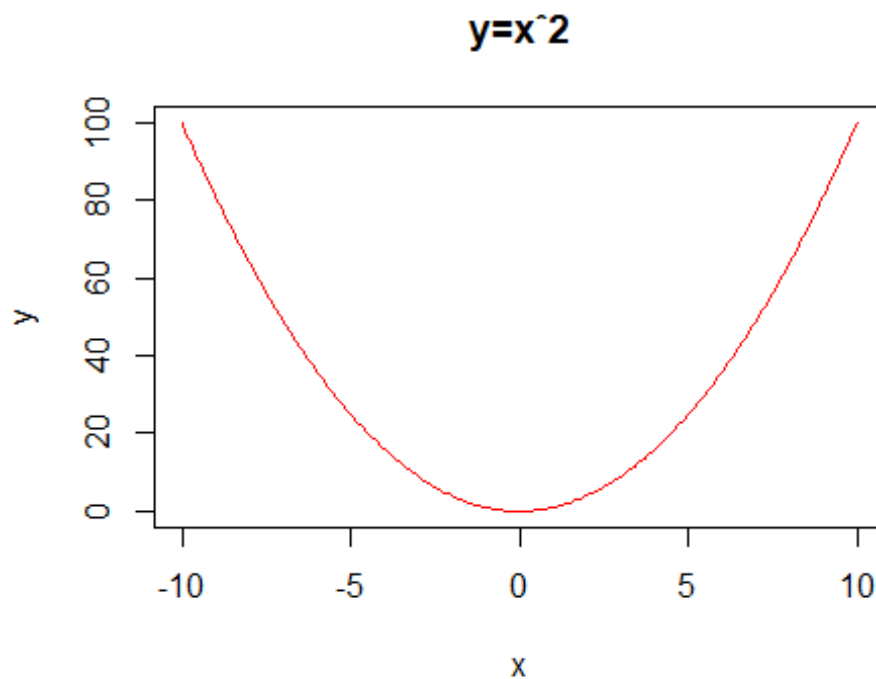
```
result=sqrt(49)
print(result)

## [1] 7

result <- exp(2)
print(result)

## [1] 7.389056

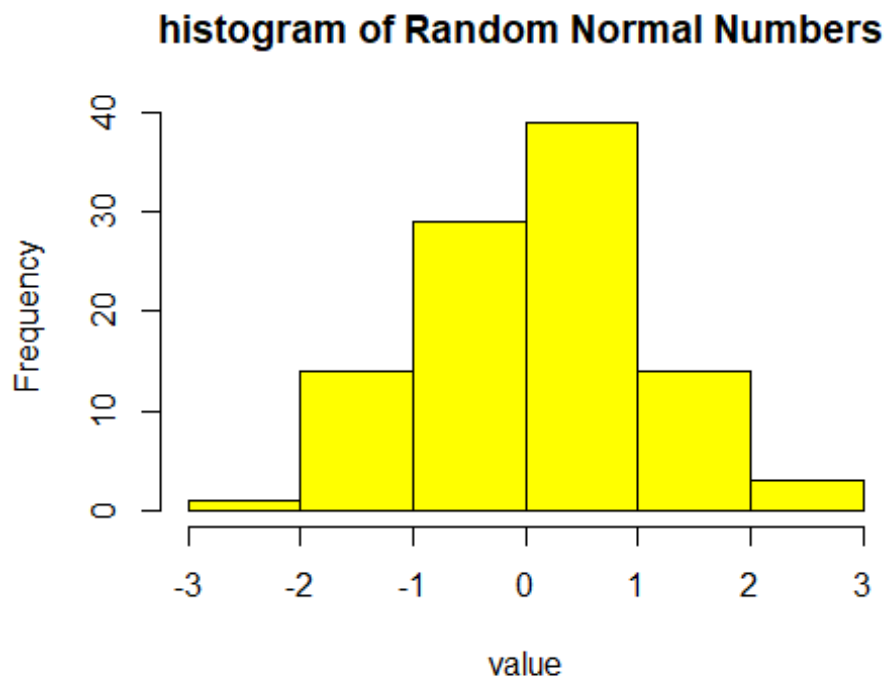
x<-seq(-10, 10, by = 0.1)
y<-(x^2)
plot(x,y,type = "l",col="red",main="y=x^2",xlab="x",ylab="y")
```



```
f <- function(x) x^2
result<-integrate(f,lower = 0,upper = 5)
print(result)

## 41.66667 with absolute error < 4.6e-13

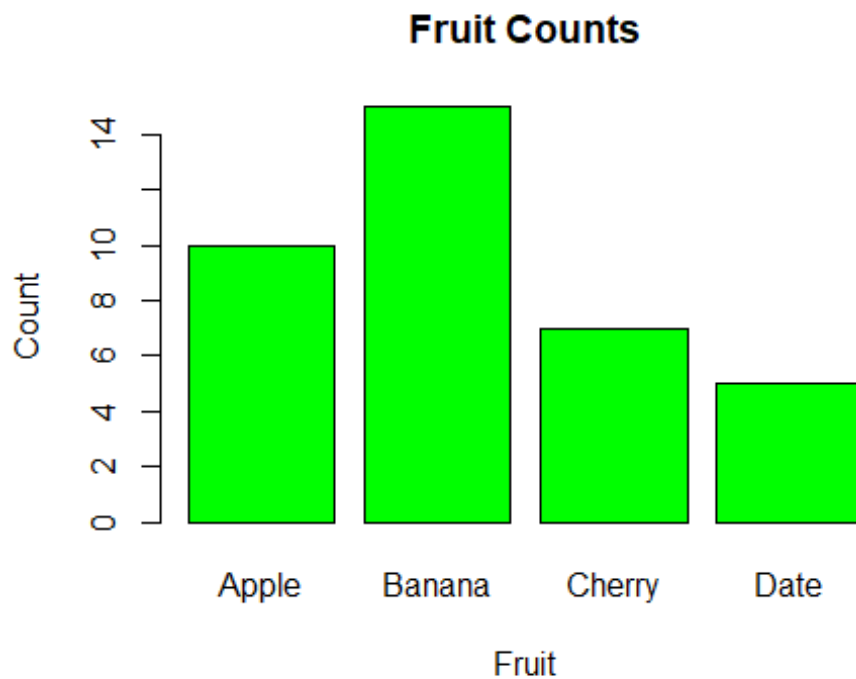
random_numbers<-rnorm(100)
hist(random_numbers,main="histogram of Random Normal Numbers ",
xlab="value",col="yellow")
```



```
result<-factorial(5)
print(result)

## [1] 120

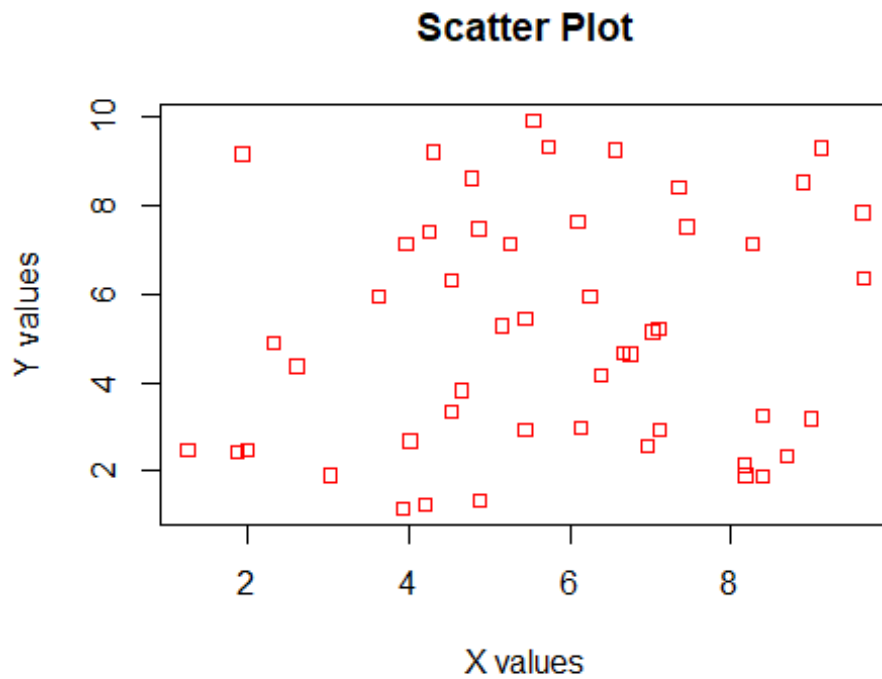
fruits <- c("Apple", "Banana", "Cherry", "Date")
counts <- c(10, 15, 7, 5)
barplot(counts, names.arg = fruits, col = "green", main = "Fruit Counts",
        xlab = "Fruit", ylab = "Count")
```



```
f <- function(x) sin(x)
result <- integrate(f, lower = 0, upper = pi)
print(result)

## 2 with absolute error < 2.2e-14

x<-runif(50, 1, 10)
y<-runif(50, 1, 10)
plot(x,y,main="Scatter Plot",xlab="X values",ylab ="Y values",col
     ="red",pch = 22)
```



```
result <- tanh(1)
print(result)

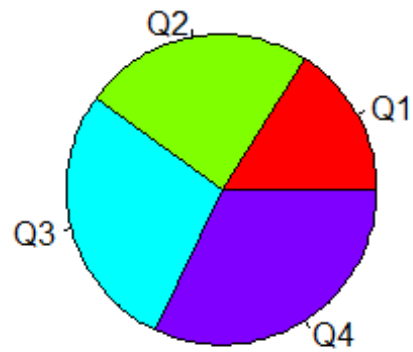
## [1] 0.7615942

result <- ceiling(5.7)
print(result)

## [1] 6

sales <- c(20, 30, 35, 40)
labels <- c("Q1", "Q2", "Q3", "Q4")
pie(sales, labels = labels, col = rainbow(4), main = "Sales Distribution")
```

Sales Distribution

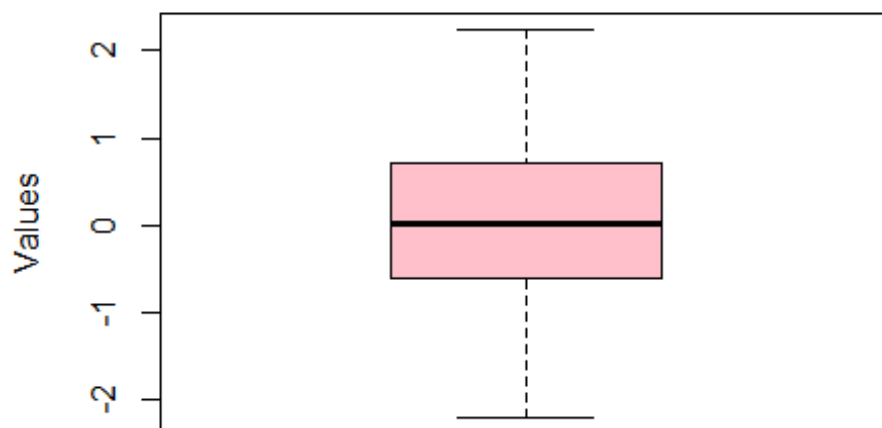


```
result <- abs(-15)
print(result)

## [1] 15

data <- rnorm(100)
boxplot(data, main = "Boxplot of Random Data", ylab = "Values", col =
        "pink")
```

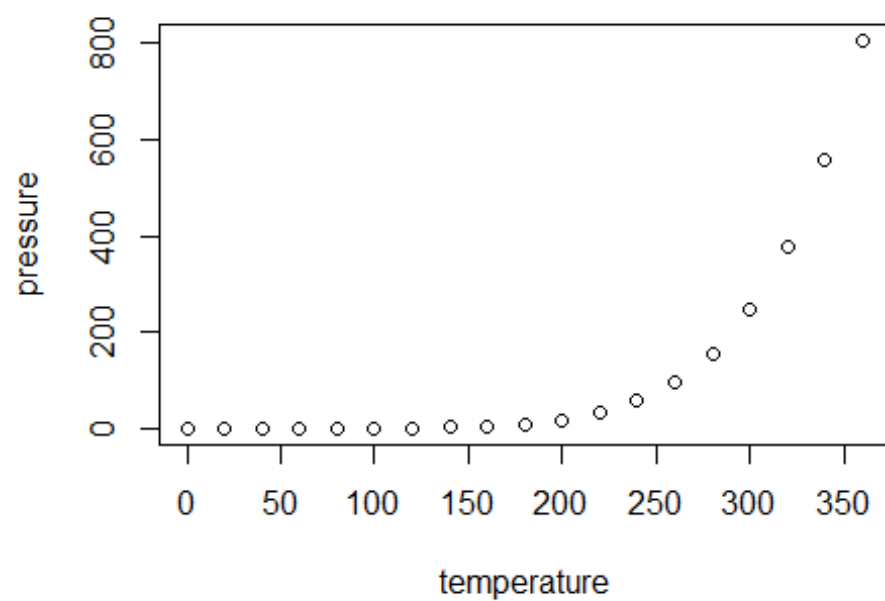
Boxplot of Random Data



```
result <- log10(1000)
print(result)
## [1] 3
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.